

# Harshil Makwana

Generative AI Engineer | AI Automation & Agentic Systems

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## SUMMARY

Computer Engineering graduate (CGPA 8.39) with hands-on experience building LLM-powered autonomous agents, RAG pipelines, and generative AI systems across two internships — shipping 10+ production AI workflows and 5 live public projects, including dual-platform publishing systems and REST APIs. Skilled in integrating LLMs with vector databases, async task pipelines, and multi-platform APIs to deliver reliable, production-grade automation. SAP Code Unnati certified (AI, Machine Learning, Deep Learning, IoT, SAP BTP) via Edunet Foundation. Focused on retrieval-augmented generation, multi-step agent orchestration, and tool-integrated LLM pipelines from prototype to cloud deployment.

## TECHNICAL SKILLS

**Generative AI & LLMs:** LLMs, Prompt Engineering, OpenAI API (GPT-4o, GPT-image), RAG Pipelines, Embeddings, Semantic Search, NLP, Hugging Face Transformers

**Agentic Systems:** Autonomous Task Agents, Multi-step Tool Orchestration, Self-Evaluation/Scoring Loops, n8n Agentic Workflows

**Backend & Deployment:** Python, FastAPI, REST APIs, Celery (Async Task Queues), Docker, Render, Railway, Netlify, Git, Linux

**ML & Data:** PyTorch, TensorFlow, Scikit-learn, Bidirectional LSTM, Neural Networks

**Databases:** PostgreSQL, PGVector (Vector Search), Supabase

**Integrations:** Telegram Bot API, Instagram Graph API, WooCommerce API, Google Docs/Sheets API, Cloudflare, Discord API, Trello API

## PROFESSIONAL EXPERIENCE

### AI Automation Engineer (Intern)

May 2026 – Present

Rohtre Media — Remote

- Built a Telegram-driven publishing agent that turns a headline and source image into a branded graphic, drafts and reviews the caption and website summary, then publishes simultaneously to a company website and Instagram — tracking each platform's publish state independently so a partial failure only retries the side that failed, never double-posting.
- Extended a content-generation agent to auto-generate and host thumbnails and publish end-to-end with zero manual writing, running autonomously 24/7 with Telegram status alerts on source volume; lifted publishing efficiency ~95%, cut writer time 5x, and held ~90% accuracy on spot-checked output.
- Automated e-commerce product onboarding (image parsing to auto-generated title/description/price) and built an n8n editorial workflow (draft to review to publish), cutting manual upload effort ~85% and publishing time ~90%.
- Designed a fully autonomous content agent that scans 20+ YouTube channels, filters relevant sports content, and drafts complete articles with a self-evaluation loop that scores each draft against an SEO rubric and regenerates up to 2 revisions before auto-selecting the best version.

### Python AI Engineer (Intern)

Nov 2025 – May 2026

Toshal Infotech Pvt. Ltd. — Surat, India

- Engineered AI-powered FastAPI microservices with Celery async task pipelines, cutting data-retrieval latency; implemented a PGVector-based RAG pipeline enabling sub-second semantic search over large document corpora.
- Integrated OpenAI LLMs for automated insight generation (summaries, keyword extraction, sentiment, topic detection) and channel-tracking alerts, reducing manual analysis time significantly; deployed and maintained services on Render.

## KEY PROJECTS

### AI-Powered Video Intelligence Platform

FastAPI · PostgreSQL · PGVector · Celery · OpenAI API · Render

- Architected a full-stack AI platform ingesting video transcripts and generating real-time insights via an LLM + RAG pipeline, achieving ~90% retrieval accuracy with a semantic search engine over large datasets.
- Built an alerts and analytics layer with automated notifications and channel-monitoring for new uploads, deployed on Render (backend) with a scalable, production-ready service architecture.

### AI News Aggregator

Python · FastAPI · OpenAI API · Cloud Deployment

- Built an end-to-end AI-powered news aggregation service from ingestion through production deployment, structuring the build across local setup, deployment configuration, and production-optimization stages.

## EDUCATION

### Vidhyadeep Institute of Engineering and Technology, GTU, Gujarat, India

B.E., Computer Engineering — CGPA : 8.39

Graduated May 2026

Bhagwan Mahavir University

Diploma, Computer Engineering — CGPA : 8.21

2020 – 2023

## CERTIFICATIONS & ACHIEVEMENTS

- SAP Code Unnati Certification – Edunet Foundation (AI, Machine Learning, Deep Learning, IoT, SAP BTP)
- 10+ production AI workflows and 5 live public projects deployed across two internships, including REST APIs, autonomous agents, and dual-platform publishing systems.